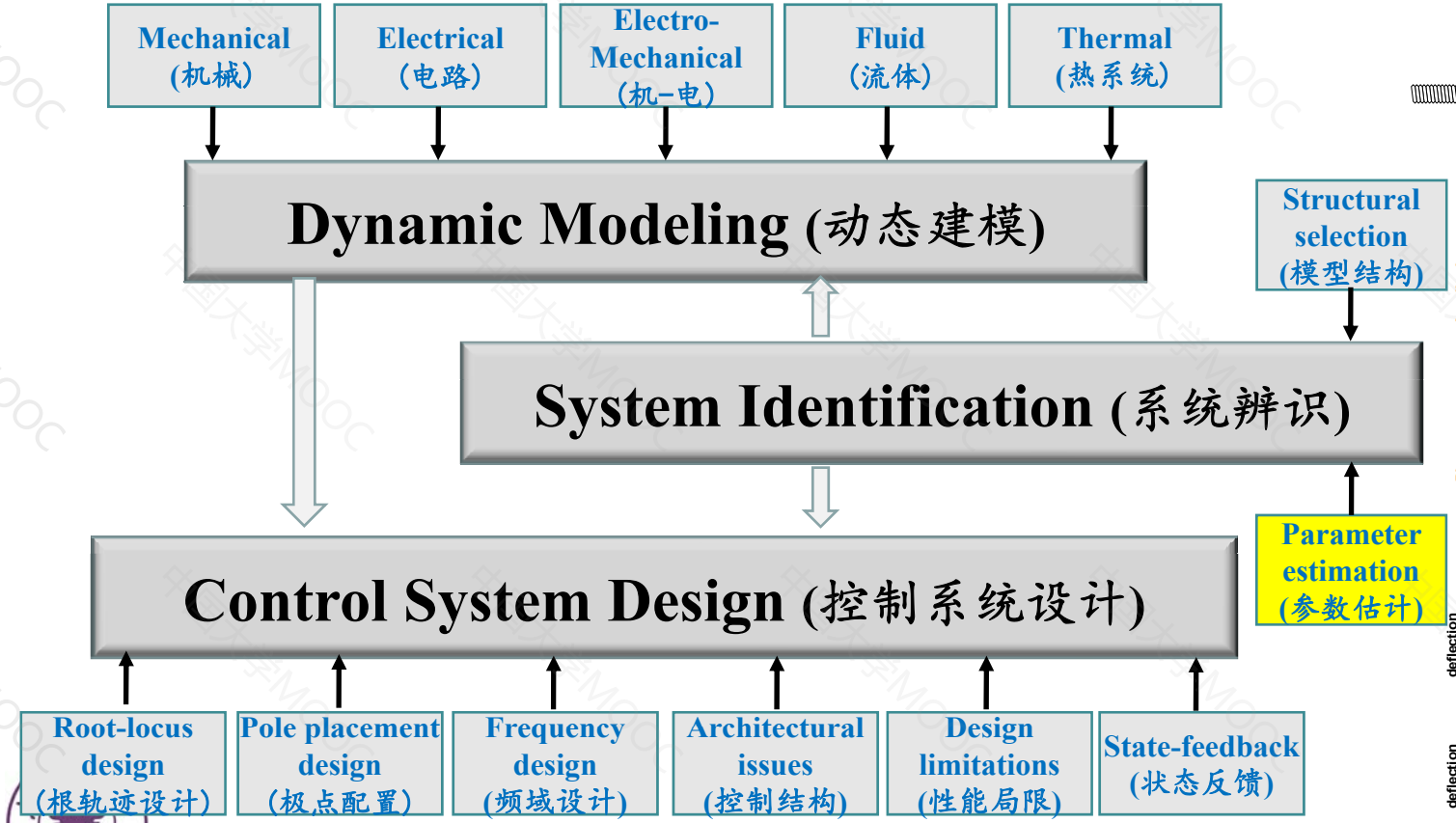
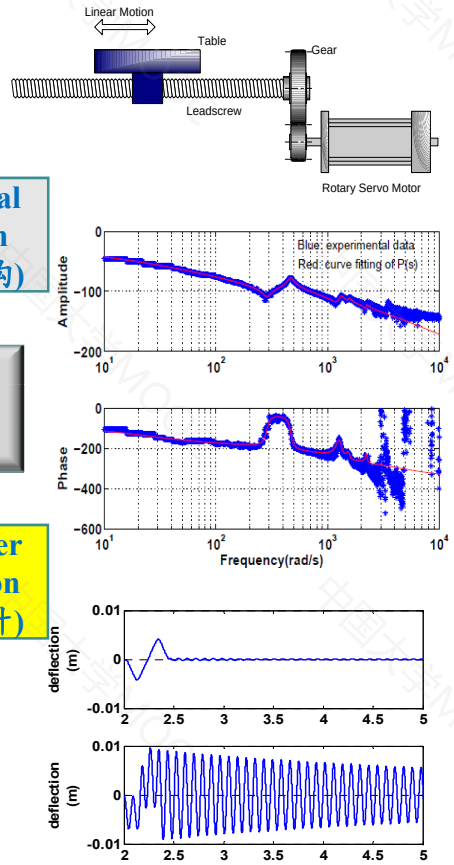
Parameter Estimation for Linear Regression Type Model Structures – Least Square Estimation

(具有线性回归类型模型结构的参数估计 – 最小二乘估计)

Main Contents of Course (课程主要内容)





Linear Regression Type Model Structures

- **Linear Regression Form** (线性回归形式)

$$y(t,\theta) = \varphi_1(t)\theta_1 + \varphi_2(t)\theta_2 + \ldots + \varphi_p(t)\theta_p = \varphi(t)^T\theta$$

✓**the predicted output vector of the model** (模型的预测输出向量)**:**

$$\hat{y} \in R^m$$

✓**parameter vector** (参数向量)**:**

$$\theta = [\theta_1(t), \theta_2(t), \ldots, \theta_p(t)]^T \in R^p$$

✓ **the vector of known basis functions' values at *t* corresponding to θ_i** (在*t*时刻相对于θ_i的已知基函数值的向量)**:**

$$\varphi_i(t) \in R^{\mathrm{m}}$$

$\varphi_i(t)$ could be nonlinear functions of the measured data $Z_t = \{U_t, Y_t\}$ ($\varphi_i(t)$可以是测量数据Z_t的非线性函数)

✓**the regressor matrix at *t*** (*t*时刻的回归矩阵)**:**

$$\varphi(t) = [\varphi_1(t), \varphi_2(t), \ldots, \varphi_p(t)]^T \in R^{p\times m}$$

EX1: Parameter Estimation of Rigid-body Motor Dynamics

- **Dynamics of Rigid-body Motor** (直线电机刚性系统的动力学)

$$\begin{cases} \dot{x}_1 = x_2 \\ M\dot{x}_2 = u - Bx_2 - F_{fn} + \Delta \\ y = x_1 \end{cases}$$

$x = [x_1, x_2]^T$ position and velocity state vector (位移和速度向量)

B equivalent viscous friction coefficient (等效阻尼摩擦系数)

$\Delta = F_d - F_r$ lumped disturbances (集中干扰)

$\bar{F}_{fn} = A_f S_f(x_2)$ modeld Coulomb friction force (建模的库仑摩擦力)

$$\Rightarrow$$

$$M\dot{x}_2 = u - Bx_2 - A_f S_f(x_2) + d$$

$d = \bar{F}_{fn} - F_{fn} + \Delta$

$\tilde{d} = d - d_n$

with

$$\theta = [\theta_1, \theta_2, \theta_3, \theta_4]$$

$$\theta_1 = M, \quad \theta_2 = B, \quad \theta_3 = A_f, \quad \theta_4 = d_n$$

$$\Rightarrow$$

$$\theta_1 \dot{x}_2 = u - \theta_2 x_2 - \theta_3 S_f(x_2) + \theta_4 + \tilde{d}$$

EX1: Parameter Estimation of Rigid-body Motor Dynamics

- **Linear Regression Form at time *t*** (时刻*t*的线性回归形式)

$$\phi(t) = [\dot{x}_2, x_2, S_f(x_2), -1]^T$$

$$\phi(t)^T \theta = u + \tilde{d}$$

$\tilde{d}$ the model structure error (模型结构误差)

$\hat{y}(t,\theta) = \varphi(t)^T \hat{\theta}$ define the predicted output (定义预测输出)

u The measured output (测量输出)

- **Prediction Error with Model Parameter Estimate** (模型参数估计的预测误差)

$$\varepsilon_\theta(i) = y(i) - \hat{y}(i,\hat{\theta}) = u(i) - \varphi(i)^T \hat{\theta}$$

$$\Rightarrow \varepsilon_\theta(i) = -\varphi(i)^T \tilde{\theta} + \tilde{d}$$

$\tilde{\theta} = \hat{\theta} - \theta$ is the parameter estimate error (参数估计误差)

Least Squares Estimation Algorithm

- **Weighted Fitting-Error-Squares Type Performance Index Function**

(加权拟合误差平方形式的性能指标函数)

$$J(t,\theta)=\sum_{i=1}^{t}\left\|R(t,i)[y(i)-\varphi(i)^T\hat{\theta}]\right\|^2=\sum_{i=1}^{t}\varepsilon_\theta(i)^T\underbrace{R(t,i)^TR(t,i)}_{W(t,i)}\varepsilon_\theta(i)$$

$$=E_\theta(t)^T\overline{W}(t)E_\theta(t)$$

$R(t,i)$ or $W(t,i)$: the weighting matrix (加权矩阵) , $\overline{W}(t)=\mathrm{diag}\{W(t,1),\ldots,W(t,t)\}$

$\varepsilon_\theta(i)=y(i)-\hat{y}(i,\theta)$: the model prediction error at time i (*t*时刻的模型预测误差)

θ: model parameter vector (模型参数向量)

- **Entire Model Error Vector** (整体模型误差向量)

$$E_\theta(t)=Y(t)-\Phi(t)\hat{\theta}$$

$$E_\theta=[\varepsilon_\theta(1)^T,\varepsilon_\theta(2)^T,\ldots,\varepsilon_\theta(t)^T]^T,\qquad Y(t)=\begin{pmatrix}y(1)\\ \vdots\\ y(t)\end{pmatrix}\in R^{tm},\quad \phi(t)=\begin{pmatrix}\varphi(1)^T\\ \vdots\\ \varphi(t)^T\end{pmatrix}\in R^{tm\times p}$$

Least Squares Estimation Algorithm

- **Necessary Condition for Solutions of Optimality** (最优解法的必要条件)

$$\left.\frac{\partial J(t,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)} = 0$$

$$J(t,\boldsymbol{\theta}) = [Y(t)-\Phi(t)\theta]^T \overline{W}(t)[Y(t)-\Phi(t)\theta]$$

$$\frac{\partial J(t,\theta)}{\partial \theta} = -\Phi(t)^T \overline{W}(t)[Y(t)-\Phi(t)\theta] = 0$$

- **Normal Equation** (一般方程) $\Downarrow$

$$\Phi(t)^T \overline{W}(t)\Phi(t)\hat{\boldsymbol{\theta}}(t) = \Phi(t)^T \overline{W}(t)Y(t)$$

$\Downarrow$ when $\Phi(t)^T \overline{W}(t)\Phi(t)$ is nonsigular (非奇异)

- **The Minimization Solution** (最小化求解)

$$\hat{\boldsymbol{\theta}}(t) = P(t)\Phi(t)^T \overline{W}(t)Y(t) = P(t)\left[\sum_{i=1}^{t} \varphi(i)W(t,i)y(i)\right]$$

- **Covariance Matrix** (协方差矩阵)

$$P(t) = \left[\Phi(t)^T \overline{W}(t)\Phi(t)\right]^{-1} = \left[\sum_{i=1}^{t} \varphi(i)W(t,i)\varphi(i)^T\right]^{-1} \in R^{p\times p}$$

Summary

- **Linear Regression Type Model Structures** (线性回归类型的模型结构)
- **Least Square Estimation (LSE) Algorithm** (最小二乘估计算法(LSE))